

Bubble Spacetime Theory — State of the Program

27 September 2026. Single source: *notes/BST_PRESENTATION_STATE_BLOCK.md* in the repository.

Where the program stands — one block, one source

*Last accuracy-synced: 2026-09-27 12:11 (Zenodo Part A on Cal Section 1003 + re-read: Fired-and-lost gains E7/E8; ‘What the geometry forces, and what it does not’; the Section A pointer). Previous: 2026-09-25 15:25 (K1923: the boundary paragraph’s parameter clause split on Cal Section 983 — V_{us} is derived given an identified input, not from the ruler). Previous: 2026-09-24 10:33 (K1921: the boundary paragraph corrected on Cal §982 — seven words; the DESI-direction sentence STRUCK as a pre-registration leak and a sign error; the four-word count 5 derived / 12 identified / 7 open / 2 input from the generator, unchanged).

The object. One rank-2 bounded symmetric domain, $D_{IV}^5 = SO(5,2)/[SO(5) \times SO(2)]$, and one measured number taken openly as the ruler. Its genus is 5; the integer 7 the program calls g is a definition — $p + q = 7$, the dimension of the defining representation of $SO(5,2)$, whose signature is the pair $(5,2)$ — not the genus — a mislabel that stood from May to September and is now swept.

Why this object — honestly. Among all irreducible bounded symmetric domains of rank at least 2, D_{IV}^5 is the unique one whose characteristic multiplicity is 3 (a theorem, checked three independent ways). That 3 is *identified* with the number of quark colours; the identification is numerical, and the bridge that would make it a mechanism is proved absent — the geometry supplies $U(1) \cdot SO(3)$, four dimensions, where colour needs $SU(3)$ ’s eight, and its triplet is self-conjugate where colour’s is not. So the selection is a fit with one measured input — the one dimensionless input *to the choice of the object*; the program has further inputs downstream (masses, a mixing corner, a CP phase), counted in the ledger — not a forcing. We do not claim zero free parameters.

Derived (mechanism proved; inputs named in the statement): the gauge-group skeleton · one fermion generation as the 16-real spinor with every hypercharge from one four-input theorem · a single-chirality positive-energy spectrum · three generations as rank+1 strata (and $m_1 = 0$) · the conserved charges · the mixing sector’s mechanism and its order — the 1–3 corner one power below the 2–3 · the Cabibbo angle, blind ($\lambda = 1/\sqrt{20}$ — derived given an identified input: the down-quark ladder’s weight, next list; the value is disfavoured: A2 fired on $K_{\mu 2}$ (Cal Sections 989/992 criterion, certified K1928 second

amendment; the $|V_{ud}|$ face past 3σ ; rescues named, not credited); FLAG 2027 decides) · time: the K-centre $SO(2)$ is the clock, the arrow is dynamical, the tick is $N_{\max} \cdot \hbar / (m_e c^2)$ · quantum mechanics on the Hardy space $H^2(D_{IV^5})$, with the Hua branching weights as the Born probabilities of the write tuple (3/7 with no table consulted). The ten Dirac–von Neumann axioms are recovered, each at its own tier, through four named posits (T2631, the ten-item row — registered 2026-09-11, Cal’s text verbatim, d7c4e474); not “10/10 derived.”

Identified (formula matches; mechanism not claimed): $\alpha^{-1} = 137$ -class expressions · the electron’s anomalous moment (our four “Selberg terms” are the four summands of Petermann–Sommerfeld’s 1957 closed form; the match is real, the derivation is not ours) · the down-quark mass ladder’s weight — $m_s/m_d = (\nu_W+1)(\nu_W+2) = 20$ with the shape derived and the weight $\nu_W = N_c$ identified, its mechanism an open gate (T2513, 2026-08-23; re-tiered K1902 and Cal Section 969, 2026-09-14) · the asymptotic-freedom coefficient $b_0 = 7$ read as g · the exact mixing values · Newton’s G : the relation between G and the electron mass is a theorem (K1673; 0.065%) that trades one dimensionful input for another and carries α inside it, so the value is a relation, not a prediction from nothing · assorted precision matches whose own null tests show small-integer expressibility is cheap.

Floored (closed by a theorem; reopenable only by a named kind of idea): Koide’s source · the mass-tower/generations mechanism · the $\ell = 2 / su(3)$ native dynamics · Λ ’s value (the closed form’s power is fixed — $p = 1$ by the Peirce weight — and at $p = 1$ the form is de Sitter’s $\Lambda \propto H^2$, a consistency with GR rather than a value; the open object is the ledger’s filling law, K1919 §7) · the nucleation survivor (no lawful initialisation vector; a distribution on the winding count) · the absence floor — no group in the geometry acts on the colour slot as colour; the door is a group from outside it (register F4, certified K1894) · the Riemann row (an attempt with a location, closed and parked) · the four-colour row (the One-Word Lemma refuted in frame; the derived lemmas stand).

The boundary — the open half (stated 2026-09-22; corrected 2026-09-24 on Cal’s cold read, Section 982, and Lyra’s collision, K1921; row-10 clause 2026-09-25, Cal Section 983). BST derives the Standard Model’s gauge skeleton, three generations at the floor, and five of its twenty-six parameters — four from one geometry and one named input (the ruler), the fifth (V_{us}) derived given an identified input (the tier table, row 10); twelve more are identified, at the precisions the tier table lists row by row, with the mechanism open. Of the remaining questions, colour’s origin is proved not to lie in the geometry’s interior (F4, a theorem), and α ’s forward reading fired and lost (E1), which closed one class of candidates; the program’s hypothesis places both on the boundary, whose physics is its open half. The first piece of that physics is the commit-

ment ledger: with the tick defined as the time to begin a commitment, Landauer's bit at the horizon temperature reproduces Λ 's exponent — by identity, since that temperature is set by H — and the closed thermostat form at the Peirce-forced power $p = 1$ is de Sitter's $\Lambda \propto H^2$: a consistency with GR that could have failed ($p = 1$ was derived on 2026-09-21 with no reference to Friedmann, a day before the identification), not a derivation of Λ 's value. The committed count obeys $d \ln N/dt = 2H$, so it grows like a comoving-sphere area. What the ledger must still supply is its filling law — how the commitment fraction changes with the scale factor; a constant fraction gives $w = 0$ and is dead on sight (register F2), and the direction of w 's evolution under a growing fraction is pre-registered to be printed blind, not stated here. No tick in the corpus produces H from the five integers; H is measured because it changes.

What the geometry forces, and what it does not (stated 2026-09-26; register Section D; K1929–K1932, Cal Sections 993–1000). D_{IV}^5 forces the kinematics — the spectra, the two-body channels with their multiplicities and selection rules, the discreteness of the interior clock — and none of the dynamics: every Minkowski mass is the ruler times a number, the couplings are identified, and at the level of the geometry the boundary's two-body spectrum is that of a free field. The channel list is forced with no free parameters; any interaction must live inside it. Hydrogen's shell structure appears one dimension down, as the minimal representation of D_{IV}^4 — recapitulated, not contained.

Fired and lost (the falsifier register's Section E — certified, published with the same ceremony as wins): E1, α is not the bare geometric vertex — the forced candidate computes to $8\pi^3/3$, closing the fifty-year volume-reading class · E2, the thermal-generations mechanism died at the order level · E3, the commit-Boltzmann ladder failed its own 5% line · E4, the sub-Tsirelson ceiling $S = 2.80624$ — refuted by Poh et al. 2015 at 41.9σ , ten years before it was registered (certified K1893, 2026-09-11). · E5, the Q^5 parity fold is a projector with no scale of its own — a forced object that fails cleanly (K1799; row certified K1894) · E6, five named series for the mixing corner, sealed by hash, all missed — the structure itself could not fail, so the negative is the five candidates (K1808/K1810; row certified K1894). · E7, $g_A = 4/\pi$ — 5.7σ from PERKEO III alone, $\geq 6\sigma$ with the bottle lifetime on every $|V_{ud}|$ route (Cal Section 984; certified 2026-09-27) · E8, “19 appears only in coloured quantities” — falsified as stated by $\Omega_\Lambda = 13/19$, a colourless 19 already on file (Cal Section 990).

Live and pre-registered (a test with its outcomes partitioned as code before the data): A9, one boost not two — the matter rest frame read from the count and redshift dipoles of distant quasars must equal the CMB boost; a replication of Wu & Xia 2026 (DESI DR1) with an independent estimator on Quiaia, frozen v1.5.2 (hash 8013d959...; seven

re-freezes before any dipole, five forced by synthetic controls); run 1 on Quaia certified 2026-09-17 10:12: Landing C — not decidable — on both samples, by the design’s own pre-target checks (profile-alternative $+7.5\sigma$; per-bin residuals $p = 0.000$; a selection systematic under the fall-back weighting, Elie 5767); the fitted vector is consistent with the CMB but the model does not describe the sky, so it is evidence neither way; A9 stays live and the next catalogue is chosen on what the C taught (K1910, Cal §976, K1911). The register’s Section A lists every live falsifier (A1–A14). A13’s dark-matter ratio 16/3 sits at its threshold (3.0σ on ACT DR6 + DESI DR2 at the published compression correlation; the chain’s own correlation is unpublished). A2 has fired on the $K_{\mu 2}$ route (above); FLAG 2027 decides.

Forbidden — any confirmed detection kills the framework: grand unification · proton decay · right-handed W or Z’ · magnetic monopoles · sterile neutrinos · a SUSY spectrum.

How to check any of this without trusting us. `python3 play/verify_bst.py` · the theorem registry (`notes/BST_AC_Theorem_Registry.md`) · the falsifier register with its fired-and-lost section · two thousand single-claim toys · every audit, retraction and correction in the open, including the ones against the program’s own auditors. The count that matters for the Standard Model row, generated from the register and never typed (`play/bst_26_tier_generator.py`, Grace, Round 142): Of the 26 primary Standard-Model parameters: 5 derived — each with its mechanism proved and its inputs named on its own statement (Cal Section 959 re-read complete, 2026-09-14) — 12 identified, 0 floored, 7 open, 2 input. (Grace’s generator, re-keyed 2026-09-14 on Cal Section 959, Elie 5755/5756, K1897, and — row 5 — K1902 with Cal Section 969; pasted from --sentence, never typed.) The July count “8 of 26 sourced clean,” carried through eleven ledger versions, is retired — its numerator and denominator came from two different lists. The honest sentence: everything reachable is derived, floored with a theorem, or published as a loss; what remains open is a short list of named doors that only new ideas — not labour — can open.